

EXP 24 :

Aim

To write a Python program using Pandas and Matplotlib to draw line charts for the financial data (Open, High, Low, Close) of Alphabet Inc. from October 3, 2016 to October 7, 2016.

Algorithm

1. Import the required libraries Pandas and Matplotlib.
2. Read the financial data from fdata.csv.
3. Convert the Date column into date format.
4. Set the Date as the x-axis.
5. Plot the Open, High, Low, and Close values as separate line charts.
6. Add labels, title, legend, and grid.
7. Display the graph.

Code

```
import pandas as pd
import matplotlib.pyplot as plt

# Read the CSV file
df = pd.read_csv("fdata.csv")

# Convert Date column to datetime
df["Date"] = pd.to_datetime(df["Date"])

# Plot financial data
plt.figure(figsize=(10, 5))

plt.plot(df["Date"], df["Open"], label="Open")
plt.plot(df["Date"], df["High"], label="High")
plt.plot(df["Date"], df["Low"], label="Low")
plt.plot(df["Date"], df["Close"], label="Close")

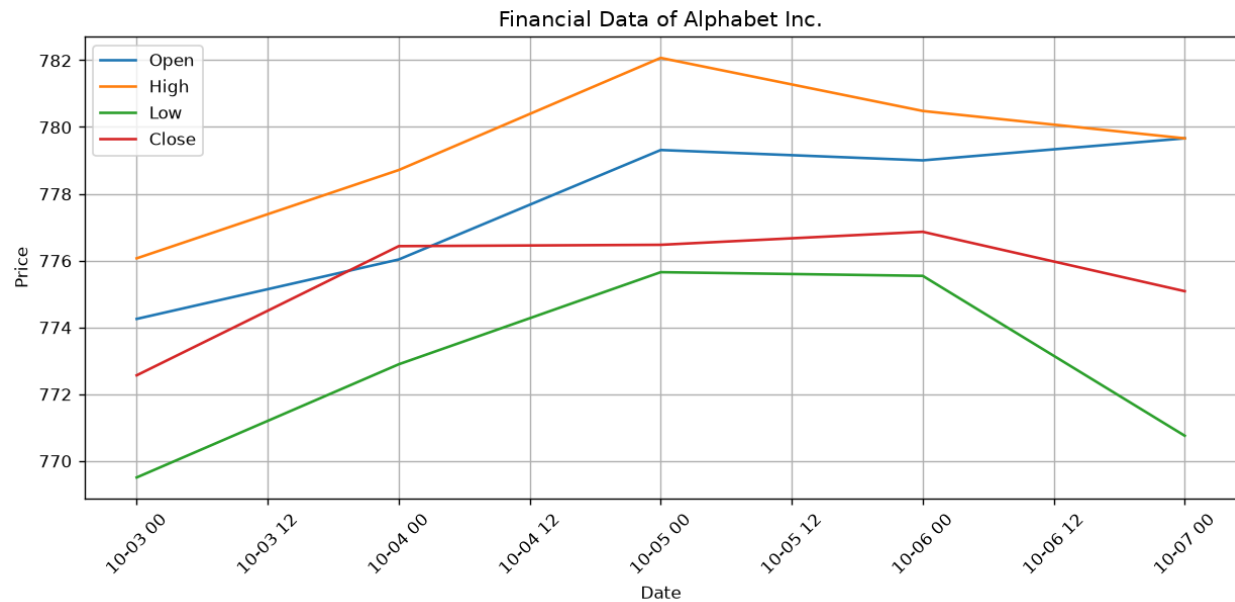
# Add labels and title
plt.xlabel("Date")
plt.ylabel("Price")
plt.title("Financial Data of Alphabet Inc.")

# Display legend and grid
plt.legend()
plt.grid(True)

# Rotate date labels
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```

Output



Result

Thus, the Python program was successfully executed to draw line charts representing the Open, High, Low, and Close financial data of Alphabet Inc. from October 3, 2016 to October 7, 2016.

EXP 25 :

Aim

To write a Python program to plot two or more lines with different colors, line widths, and suitable legends using Matplotlib.

Algorithm

1. Import matplotlib.pyplot.
2. Define the x and y values for two lines.
3. Plot the first line with a blue color and width 3.
4. Plot the second line with a red color and width 5.
5. Add legends, title, and axis labels.
6. Display the graph.

Code

```
import matplotlib.pyplot as plt

x = [10, 20, 30]

y1 = [20, 40, 10]
y2 = [40, 10, 30]

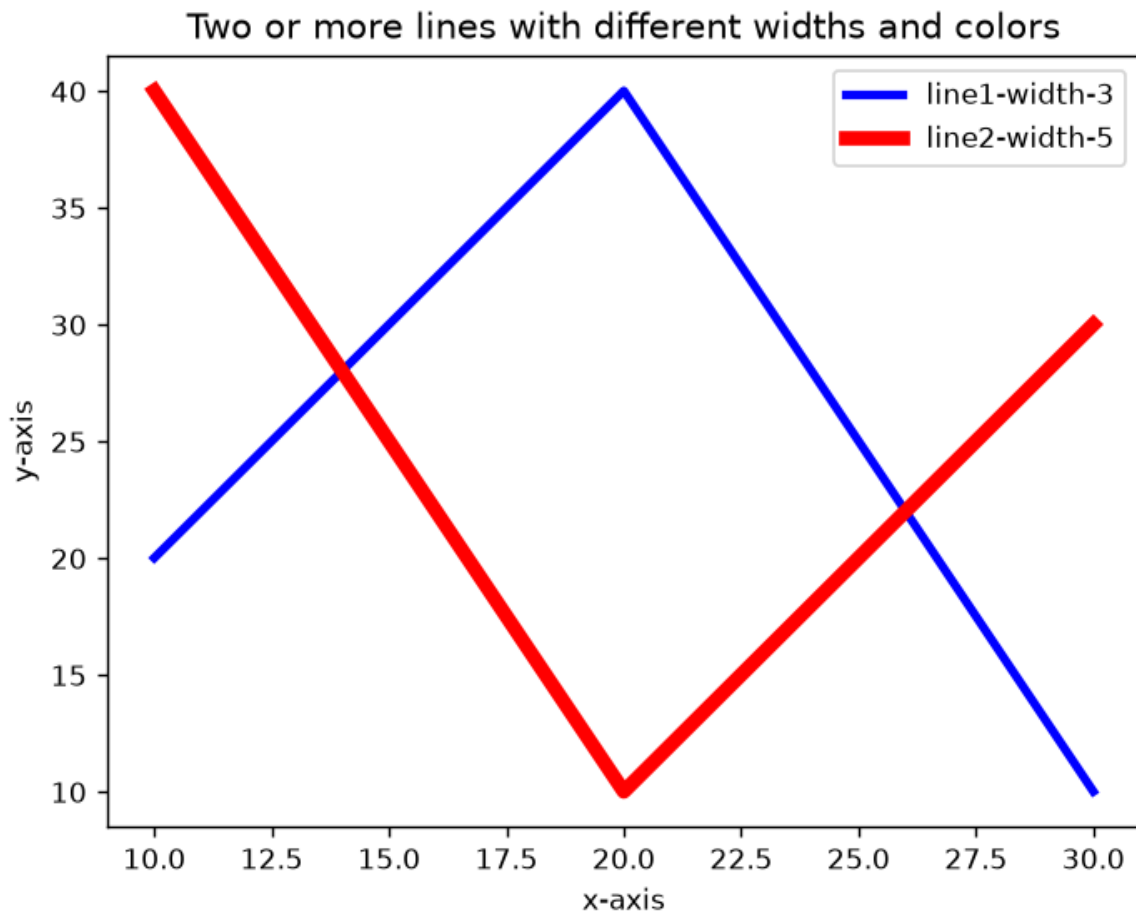
plt.plot(x, y1, color='blue', linewidth=3,
         label='line1-width-3')

plt.plot(x, y2, color='red', linewidth=5,
         label='line2-width-5')

plt.xlabel('x-axis')
plt.ylabel('y-axis')
plt.title('Two or more lines with different widths and colors')

plt.legend()
plt.show()
|
```

Output



Result

Thus, the Python program was successfully executed to plot two lines with different colors, widths, and legends.

EXP 26 :

Aim

To write a Python program to create multiple plots in a single figure using Matplotlib.

Algorithm

1. Import matplotlib.pyplot.
2. Create a figure with multiple subplot areas.
3. Plot different data in each subplot.
4. Add suitable titles to the plots.
5. Display all plots using plt.show().

Code

```
import matplotlib.pyplot as plt

# Create multiple plots
plt.figure(figsize=(8, 6))

# First plot
plt.subplot(2, 3, 1)
plt.plot([1, 2, 3, 4], [10, 20, 15, 25])
plt.title("Plot 1")

# Second plot
plt.subplot(2, 3, 2)
plt.plot([1, 2, 3, 4], [5, 15, 10, 20])
plt.title("Plot 2")

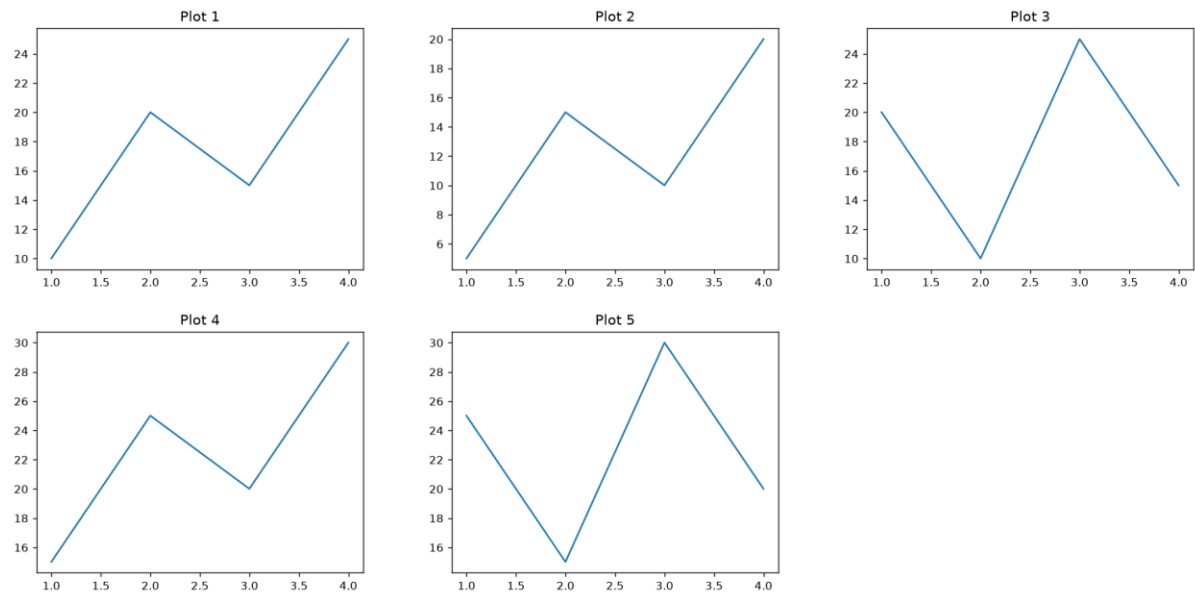
# Third plot
plt.subplot(2, 3, 3)
plt.plot([1, 2, 3, 4], [20, 10, 25, 15])
plt.title("Plot 3")

# Fourth plot
plt.subplot(2, 3, 4)
plt.plot([1, 2, 3, 4], [15, 25, 20, 30])
plt.title("Plot 4")

# Fifth plot
plt.subplot(2, 3, 5)
plt.plot([1, 2, 3, 4], [25, 15, 30, 20])
plt.title("Plot 5")

plt.tight_layout()
plt.show()
```

Output



Result

Thus, the Python program was successfully executed to create multiple plots in a single figure using Matplotlib.